

SEMICON India Hackathon 2026

KLA Track

AI-Based Restoration of Degraded Images for Semiconductor Inspection

Team: Phantom Protocol

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GitHub: <https://github.com/lords-coder/semicon-restoration>

Demo Video: [INSERT YOUR YOUTUBE/LINK HERE]

1. Problem Statement

Semiconductor wafers undergo visual inspection at multiple stages of manufacturing. These inspection images often suffer from noise introduced by imaging sensors, low resolution due to hardware limitations, and degradation from data compression and transmission. Manual inspection of degraded images is time-consuming, error-prone, and does not scale with modern semiconductor production volumes.

Objective: Build an AI model that takes degraded 128x128 noisy grayscale images and restores them to clean 256x256 high-resolution outputs, improving both visual quality and defect detection accuracy.

2. Our Solution: RestoreNet

We designed RestoreNet, a lightweight convolutional neural network for single-image super-resolution and denoising, specifically tailored for semiconductor inspection images.

Architecture

```
Input (1x128x128)
|
Conv2d(1 -> 64, 3x3)
|
8x ResidualBlock (Conv -> LeakyReLU -> Conv + Skip)
|
Channel Attention (AvgPool -> FC -> Sigmoid -> Scale)
|
Conv2d(64 -> 64, 3x3) + Skip from first conv
|
PixelShuffle 2x: Conv2d(64 -> 256) -> Reshape -> LeakyReLU
|
Conv2d(64 -> 1, 3x3)
|
Output (1x256x256)
```

3. Key Components

Component	Purpose
Residual Blocks (x8)	Deep feature extraction with skip connections
Channel Attention	Dynamic feature channel weighting (reduction=16)
PixelShuffle (2x)	Learnable sub-pixel upsampling 128 -> 256
Skip Connection	Preserves low-level features from first conv
LeakyReLU (0.2)	Non-linear activation, avoids dead neurons

4. Training Details

Parameter	Value
Framework	PyTorch 2.12+ (CUDA 12.8)
GPU	NVIDIA GeForce RTX 5060 (Laptop)
Epochs	450
Batch Size	16
Loss Function	L1 Loss (Mean Absolute Error)
Optimizer	Adam
Learning Rate	2e-4
Data Augmentation	Random H-flip, V-flip, 90-deg rotation
Checkpointing	Every 50 epochs

5. Results

Metric	Value
Training Image Pairs	3,200
Test Images	400
Training Epochs	450
PSNR (Training)	~74+ dB
Inference Speed	<50ms per image (GPU)
Model Size	~1.2M parameters

6. How to Use

Installation

```
git clone https://github.com/lords-coder/semicon-restoration.git
cd semicon-restoration
pip install -r requirements.txt
```

Restore Images

```
python restore_menu.py
```

- Option 1: Single image restoration (.npy, .png, .jpg)
- Option 2: Batch folder restoration
- Outputs: restored .png + side-by-side comparison

Batch Evaluation

```
python evaluation.py --input_dir test_noisy_lr/NoisyLR --output_dir output/
```

7. Dataset

Download from: Google Drive - KLA Dataset

<https://drive.google.com/drive/folders/1VKiFW-kDk9-q5XRPu3nrl08OM94EwzV6>

Dataset	Format / Size / Count
Ground Truth (GT)	.npy 256x256 uint8 3,200 images
NoisyLR (Train)	.npy 128x128 uint8 3,200 images
Test NoisyLR	.npy 128x128 uint8 400 images

8. Links

GitHub Repository: <https://github.com/lords-coder/semicon-restoration>

Demo Video: [\[INSERT YOUR YOUTUBE/LINK HERE\]](#)

Hackathon Portal: <https://hackathon2026.i4c.in/>

KLA Webinar: <https://youtu.be/RMSDaviTOlw>